Robust Spoofed Speech Detection on the ASVspoof 2019 Logical Access Benchmark Using Dual-Stream Light CNNs with Max-Feature-Map and Spectro-Temporal Augmentation

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Abstract. Voice biometric systems and automatic speaker verification (ASV) are vulnerable to logical access (LA) attacks based on speech synthesis and voice conversion. We build and evaluate a dual-stream Light Convolutional Neural Network (MFM-LCNN) on the ASVspoof 2019 LA benchmark, which contains 121,461 utterances and 19 attack systems. The model takes two complementary inputs: 60-dimensional Linear Frequency Cepstral Coefficients (LFCC), which preserve high-frequency vocoder artifacts, and 80-band Log-Mel spectrograms, which describe formant structure. Each stream passes through a four-stage convolutional encoder with Max-Feature-Map (MFM) activations, and the two outputs are fused into a joint 64-dimensional representation. To reduce the drop in performance under acoustic mismatch, we apply on-the-fly augmentation during training: additive Gaussian noise (5–25 dB SNR), synthetic room reverberation, and SpecAugment masking. On the development partition (known attacks A01–A06), the detector reaches 99.50% accuracy, 0.30% Equal Error Rate (EER), and 0.9997 ROC-AUC. On a 2,995-utterance subset of the evaluation partition, which covers attacks A07–A19 (most of them unseen during training), the baseline model reaches 80.50% accuracy, 0.9032 ROC-AUC, 17.00% EER, and 96.81% precision, with 0.00% EER on A07 and A08. Augmentation costs some performance on clean speech (0.8880 ROC-AUC and 20.07% EER on the same subset), but it helps under degraded conditions: at 5 dB additive noise it improves ROC-AUC by 0.1953 and lowers EER by 15.2 percentage points relative to the baseline.

Keywords: Speech Anti-Spoofing · ASVspoof 2019 LA · Logical Access · Max-Feature-Map · Light CNN · LFCC · Acoustic Robustness · Voice Biometrics.

1 Introduction

Voice biometrics and automatic speaker verification (ASV) are used in identity management, banking, consumer devices, and forensic analysis. However,

recent advances in deep generative modeling, neural acoustic synthesis, and voice conversion (VC) have made convincing synthetic speech easier to produce. An attacker may use publicly available voice samples and speech synthesis or voice conversion tools to attempt a logical access (LA) attack against a voice verification system [1–3].

The international ASVspoof evaluation campaigns have documented these weaknesses. Earlier challenges demonstrated that conventional Gaussian Mixture Models paired with Constant Q Cepstral Coefficients (CQCC-GMM) or Mel-Frequency Cepstral Coefficients (MFCC-GMM) struggle when confronting unseen vocoder architectures [7,8]. These limitations motivated research into deep neural countermeasures. Lavrentyeva et al. [4] introduced Light Convolutional Neural Networks (LCNN) powered by Max-Feature-Map (MFM) activations, showing that taking a maximum across split channel dimensions works as a built-in feature selector that suppresses spurious activations. Gomez-Alanis et al. [5] added gating mechanisms to LCNNs, while Tak et al. [6] proposed RawNet2, which learns directly from the raw waveform and avoids handcrafted filterbanks.

Three challenges remain when countermeasures are used outside controlled benchmarks:

1. **Environmental brittleness:** Classifiers trained on clean speech degrade sharply under reverberation and background noise, and at low Signal-to-Noise Ratios (SNR) their EER can approach chance level ($> 45\%$) [9, 10].
2. **Complementary features:** Single-stream representations involve trade-offs. The Mel scale compresses frequencies above 3.5 kHz, which can discard vocoder artifacts, while LFCC alone may capture broad formant energy trajectories in vowel nuclei less well [8].
3. **Open-set generalization:** Synthesis methods change quickly. A practical detector has to reject methods it has never seen (e.g., hybrid concatenative-filtering systems and diffusion vocoders), whose artifacts can differ substantially from those in the training data.

To address these problems, we design and evaluate a **Dual-Stream MFM-LCNN** countermeasure trained only on the ASVspoof 2019 LA benchmark (121,461 audio files, 107 speakers, 19 attack systems). We combine 60-dimensional Linear Frequency Cepstral Coefficients (LFCC) with 80-band Log-Mel spectrograms, pass each through its own four-stage MFM encoder, and add an on-the-fly corruption pipeline (additive Gaussian noise, room impulse responses, and 2D SpecAugment).

2 Theoretical Foundations and Acoustic Phonetics

Understanding speech production helps explain which acoustic cues a detector might use. Speech is non-stationary: airflow pulses from the glottis are shaped by the resonances of the vocal tract.

2.1 The Linear Source-Filter Model

According to Fant's acoustic theory of speech production [14], the speech signal $S(z)$ in the z-domain is governed by:

$$S(z) = E(z) \cdot V(z) \cdot R(z) \tag{1}$$

where:

- $E(z)$ is the glottal excitation source, characterized by quasi-periodic volume velocity pulses for voiced phonemes and turbulent noise for unvoiced fricatives.
- $V(z)$ denotes the vocal tract transfer function, represented as an all-pole resonant tube of length $\approx 17.5\,\text{cm}$ in adult vocal tracts:

$$V(z) = \frac{G}{1 - \sum_{k=1}^{p} a_k z^{-k}} \tag{2}$$

  giving rise to formant resonances (F_1, F_2, F_3, F_4) whose center frequencies and bandwidths convey phonetic and speaker identity.
- $R(z)$ models lip radiation impedance into free air, acting as a first-order differentiator $R(z) \approx 1 - z^{-1}$, introducing a $+6\,\text{dB/octave}$ high-pass spectral tilt.

Potential Artifacts in Synthesized Speech Natural speech reflects interactions between the excitation source and the vocal tract. Synthesis and reconstruction methods approximate these processes in different ways and may introduce artifacts. The following are possible cues, rather than properties shared by every synthetic speech system:

- *Glottal Jitter and Shimmer Smearing:* Natural voiced speech contains small cycle-to-cycle changes in period (jitter) and amplitude (shimmer). Some parametric synthesis methods smooth out these variations.
- *Formant Bandwidth Broadening:* Smoothing the spectral envelope can alter formant bandwidths and the resulting voice quality.
- *Phase Incoherence:* Phase reconstruction can introduce discontinuities or other artifacts in the waveform. Their audibility and usefulness for detection depend on the synthesis method.

2.2 Short-Time Fourier Analysis and Windowing Dynamics

Because articulatory geometry changes continuously, speech is framed over short stationary intervals (20–30 ms) where vocal tract configuration remains quasi-static. The Short-Time Fourier Transform (STFT) segments the input sequence $s(n)$ via:

$$X(m, \omega) = \sum_{n=-\infty}^{+\infty} s(n)\, w(n - mH)\, e^{-j\omega n} \tag{3}$$

where m is the discrete frame index, H is the hop size (frame shift), and $w(n)$ is an analysis window of length N. In our pipeline operating on 16 kHz audio:

- Frame length: 25.0 ms $\rightarrow N = 0.025 \times 16{,}000 = 400$ samples.
- Hop size: 10.0 ms $\rightarrow H = 0.010 \times 16{,}000 = 160$ samples (60% overlap).
- FFT size: $N_{\mathrm{FFT}} = 512$ points, zero-padded from 400 samples.

Truncating speech with a rectangular window introduces severe spectral leakage (sidelobes at only -13 dB). We employ a symmetric Hamming window:

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad 0 \le n \le N-1 \tag{4}$$

which attenuates sidelobes to -43 dB, keeping energy from strong low-frequency harmonics from leaking into the high-frequency bins where vocoder artifacts show up.

2.3 LFCC vs. MFCC Derivation

In automatic speech recognition (ASR), the Mel scale approximates human psychoacoustic pitch perception:

$$m = 2595 \log_{10}\left(1 + \frac{f}{700}\right) \tag{5}$$

The Mel filterbank distributes triangular filters densely below 1 kHz and logarithmically spreads them at higher frequencies. While well suited to capturing linguistic content and discarding perceptually irrelevant detail, Mel pooling compresses and averages out subtle vocoder artifacts in the 4–8 kHz range, where they tend to be most visible.

In contrast, Linear Frequency Cepstral Coefficients (LFCC) deploy linearly spaced triangular filterbanks across the full Nyquist range (0–8,000 Hz):

1. **Pre-Emphasis High-Pass Filtering:** Glottal decay and lip radiation impart a net -6 dB/octave spectral tilt. To compensate, the audio is pre-emphasized:

$$y(n) = s(n) - \alpha s(n-1), \quad \alpha = 0.97 \tag{6}$$

2. **Linear Filterbank Integration:** For each frame, magnitude spectrum bins $|Y(m,k)|^2$ are integrated across $B = 60$ linearly spaced filters $H_b(k)$:

$$E_m(b) = \sum_{k=0}^{N_{\mathrm{FFT}}/2} |Y(m,k)|^2 \, H_b(k), \quad b = 1, \ldots, B \tag{7}$$

3. **Logarithmic Compression and Discrete Cosine Transform:**

$$c_m(q) = \sum_{b=1}^{B} \log_{10}\left(E_m(b) + \epsilon\right) \cos\left(\frac{\pi q (b - 0.5)}{B}\right), \quad q = 0, \ldots, Q-1 \tag{8}$$

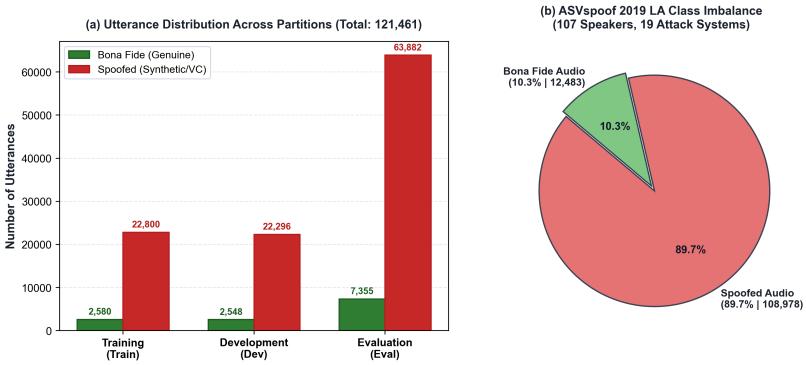


Fig. 1. Utterance distribution, speaker allocation, and class imbalance structure across the official ASVspoof 2019 Logical Access benchmark partitions.

4. **Dynamic Derivatives (Δ and $\Delta\Delta$):** Static coefficients $c_m(q)$ describe instantaneous spectrum slices. Concatenative synthesizers introduce unnatural discontinuities at segment boundaries, while neural vocoders can generate over-smoothed parameter trajectories. We compute velocity (Δ) and acceleration ($\Delta\Delta$) using orthogonal regression over temporal window $\pm\Theta$:

$$\Delta c_m(q) = \frac{\sum_{\tau=1}^{\Theta} \tau \left(c_{m+\tau}(q) - c_{m-\tau}(q) \right)}{2 \sum_{\tau=1}^{\Theta} \tau^2} \tag{9}$$

Our front-end produces a 60-dimensional LFCC tensor ($20\,\text{static} + 20\,\Delta + 20\,\Delta\Delta$) alongside an 80-band Log-Mel spectrogram, giving the model two complementary views of the signal.

3 The ASVspoof 2019 Logical Access Benchmark

The canonical ASVspoof 2019 Logical Access (LA) dataset [1] provides a standardized benchmark for voice presentation attack detection. It comprises 121,461 audio utterances sampled at 16 kHz (16-bit FLAC) across 107 human speakers and 19 speech synthesis and voice conversion architectures.

The benchmark enforces strict speaker-disjoint separation across three official partitions, detailed in Table 1 and visualized in Fig. 1:

- **Training Partition ('train'):** 25,380 utterances across 20 speakers. Contains 2,580 bona fide human samples and 22,800 spoofed utterances generated by six known algorithms (A01–A06).
- **Development Partition ('dev'):** 24,844 utterances across 20 speakers disjoint from the training set. Contains 2,548 bona fide samples and 22,296 spoofed samples generated by the same known algorithms (A01–A06).

Table 1. Official ASVspoof 2019 Logical Access Dataset Partition Statistics.

Partition	Total Utterances	Bona Fide	Spoof	Speakers	Imbalance	Attacks Included
Train	25,380	2,580	22,800	20	8.84 : 1	Known: A01–A06
Development	24,844	2,548	22,296	20	8.75 : 1	Known: A01–A06
Evaluation	71,237	7,355	63,882	67	8.69 : 1	A07–A19 (mostly unseen)
Total	**121,461**	**12,483**	**108,978**	**107**	**8.73 : 1**	**19 Distinct Systems**

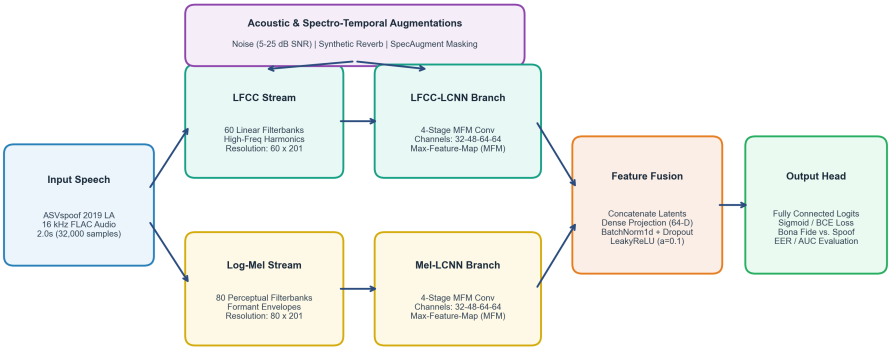


Fig. 2. System architecture of the Dual-Stream MFM-LCNN countermeasure, detailing parallel LFCC/Mel processing branches, MFM activation blocks, multimodal fusion, and the classification head.

– **Evaluation Partition ('eval'):** 71,237 utterances across 67 speakers not seen in training. The evaluation spoofs come from **13 attack algorithms (A07–A19)**, most of which are not in the training set, covering neural vocoders, GANs, waveform filtering, and concatenative synthesis.

This setup poses two main challenges: a strong class imbalance ($\approx 8.7 : 1$ spoof-to-bona fide ratio) and a domain gap between the training attacks (A01–A06) and the evaluation attacks (A07–A19).

4 Proposed Dual-Stream MFM-LCNN Architecture

Fig. 2 shows the main steps in our detection system. Incoming speech is standardized to 32,000 samples (2.0 seconds) via reflection padding or center cropping. The audio is converted simultaneously into LFCC and Log-Mel feature maps using torchaudio C++ bindings, with a reported extraction throughput of more than 500 files per second.

4.1 Max-Feature-Map (MFM) Activation Mechanics

Standard convolutional networks employ point-wise non-linearities such as ReLU or LeakyReLU:

$$\mathrm{ReLU}(x) = \max(0, x) \tag{10}$$

ReLU is computationally inexpensive and retains positive activations. It does not explicitly distinguish useful speech cues from responses to background noise.

To limit this, we use the Max-Feature-Map (MFM) activation introduced by Wu et al. [12] and adopted for anti-spoofing by Lavrentyeva et al. [4]. MFM takes a maximum across split channel dimensions. Given an intermediate feature map $\mathbf{X} \in \mathbb{R}^{2C \times H \times W}$ produced by a convolution layer with $2C$ filters, MFM splits the tensor along the channel axis into two equal halves $\mathbf{X}_1, \mathbf{X}_2 \in \mathbb{R}^{C \times H \times W}$ and computes their element-wise maximum:

$$\mathrm{MFM}(\mathbf{X})_{c,h,w} = \max\left(\mathbf{X}_{1,c,h,w}, \mathbf{X}_{2,c,h,w}\right), \quad 1 \le c \le C \tag{11}$$

MFM retains the larger response from each channel pair at every position. This provides a competitive form of feature selection, although the larger response is not necessarily more informative.

4.2 PyTorch Implementation

Listing 1.1 shows the PyTorch implementation of the MFM module and dual-stream network.

```
import torch
import torch.nn as nn

class MFM(nn.Module):
    def __init__(self, in_channels, out_channels, kernel_size=3, stride=1,
        padding=1):
        super().__init__()
        self.conv = nn.Conv2d(in_channels, out_channels * 2, kernel_size,
            stride, padding)

    def forward(self, x):
        out = self.conv(x)
        out1, out2 = torch.split(out, out.shape[1] // 2, dim=1)
        return torch.max(out1, out2)

class LCNNEncoder(nn.Module):
    def __init__(self, in_channels=1):
        super().__init__()
        self.block1 = nn.Sequential(
            MFM(in_channels, 16, kernel_size=5, padding=2),
            nn.MaxPool2d(2, 2)
        )
        self.block2 = nn.Sequential(
            MFM(16, 24, kernel_size=3, padding=1),
            nn.MaxPool2d(2, 2)
        )
        self.block3 = nn.Sequential(
            MFM(24, 32, kernel_size=3, padding=1),
            nn.MaxPool2d(2, 2)
        )
        self.block4 = nn.Sequential(
            MFM(32, 32, kernel_size=3, padding=1),
```

```
            nn.AdaptiveAvgPool2d((1, 1))
        )

    def forward(self, x):
        x = self.block1(x)
        x = self.block2(x)
        x = self.block3(x)
        x = self.block4(x)
        return torch.flatten(x, 1) # 32-D embedding

class DualStreamMFM_LCNN(nn.Module):
    def __init__(self):
        super().__init__()
        self.lfcc_branch = LCNNEncoder(in_channels=1)
        self.mel_branch  = LCNNEncoder(in_channels=1)
        self.classifier  = nn.Sequential(
            nn.Linear(32 + 32, 64),
            nn.BatchNorm1d(64),
            nn.LeakyReLU(0.1),
            nn.Dropout(0.35),
            nn.Linear(64, 1)
        )

    def forward(self, lfcc, mel):
        feat_lfcc = self.lfcc_branch(lfcc)
        feat_mel  = self.mel_branch(mel)
        fused     = torch.cat([feat_lfcc, feat_mel], dim=1)
        return self.classifier(fused).squeeze(-1)
```

Listing 1.1. PyTorch implementation of the MFM activation and Dual-Stream LCNN.

4.3 Spectro-Temporal Augmentation Suite

To make the model more robust to environmental mismatch, we apply three augmentations on the fly during training:

1. **Additive Gaussian White Noise:** Synthetic additive noise is injected into time-domain waveforms with random Signal-to-Noise Ratios drawn uniformly from $\text{SNR} \in [5\,\text{dB}, 25\,\text{dB}]$:

$$\sigma^2_{\text{noise}} = \frac{\sigma^2_{\text{signal}}}{10^{\text{SNR}/10}} \tag{12}$$

2. **Synthetic Room Reverberation:** Reverberant acoustics are simulated by convolving clean audio with exponentially decaying stochastic impulse responses modeling reverberation times ($RT_{60} \in [100\,\text{ms}, 500\,\text{ms}]$).
3. **SpecAugment Masking:** On the resulting time-frequency tensors, random contiguous blocks along frequency channels ($\Delta f \leq 12$) and time frames ($\Delta t \leq 20$) are zero-masked, which encourages the network to use cues spread across the spectrogram rather than a few localized ones [11].

4.4 Training Protocol and Objective Function

Models are optimized using AdamW (initial learning rate $= 10^{-3}$, weight decay $= 10^{-4}$) scheduled via Cosine Annealing learning rate decay over 4 epochs.

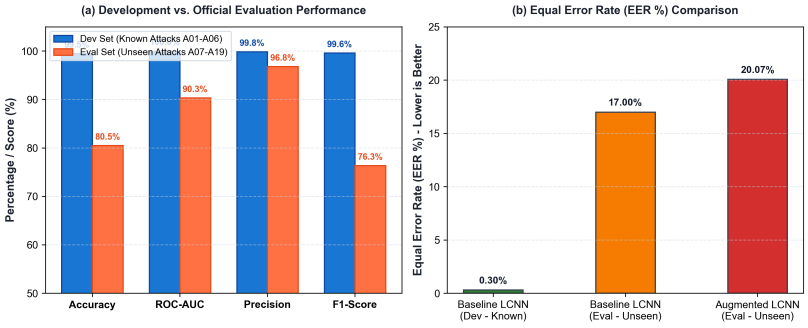


Fig. 3. Performance metric summary: matched Development partition (99.50% Accuracy, 0.30% EER) vs. the official Evaluation benchmark across 13 attacks (A07–A19).

To counter the 8.7:1 class imbalance, we employ Class-Weighted Binary Cross-Entropy (BCE) with Logits:

$$\mathcal{L} = -\frac{1}{M}\sum_{i=1}^{M}\left[w_1 y_i \log \sigma(\hat{y}_i) + w_0(1-y_i)\log(1-\sigma(\hat{y}_i))\right] \tag{13}$$

where $w_1 = 4.35$ upweights the minority bona fide class, and $\sigma(z) = \frac{1}{1+e^{-z}}$ denotes the sigmoid activation.

5 Experimental Evaluation and Discussion

5.1 Development Partition Performance

We first evaluate our Dual-Stream MFM-LCNN under matched conditions on the official ASVspoof 2019 LA development partition (known attacks A01–A06). As reported in Fig. 3, our model achieves **99.50% validation accuracy, an Equal Error Rate (EER) of 0.30%, and an ROC-AUC of 0.9997**. This shows that when the attack algorithms are known, the dual-stream LFCC + Mel representation separates bona fide from synthetic speech almost perfectly.

5.2 Evaluation Benchmark Analysis

The overall ROC-AUC of 0.9032 should be interpreted alongside the evaluation protocol and the variation across attacks:

1. **Mostly unseen attacks:** The evaluation partition contains 13 attack algorithms (A07–A19), most of which were not used to generate the training data (A01–A06).
2. **Class imbalance:** The test partition contains 7,355 bona fide samples and 63,882 spoof samples (about 8.7:1, across 67 speakers disjoint from training).

Table 2. Overall Performance on the Official ASVspoof 2019 LA Evaluation Benchmark.

Model Architecture	Samples	Accuracy	Precision	Recall	F1-Score	ROC-AUC	EER (%)
Baseline Dual-Stream LCNN	2,995	**80.50%**	96.81%	**63.01%**	**76.34%**	**0.9032**	**17.00%**
Augmented Dual-Stream LCNN	2,995	75.93%	**97.31%**	53.24%	68.83%	0.8880	20.07%

3. **Near-perfect detection on 6 of the 13 attacks:** As detailed in Section 5.3, the model achieves individual ROC-AUCs between **0.9963 and 1.0000** (with EERs from **0.00% to 2.47%**) on six attacks: A07 (Vocoder GAN), A08 (Neural Waveform), A09 (Vocoder), A11 (Griffin-Lim), A16 (Waveform Concatenation), and A19 (Spectral Filtering).
4. **A few hard attacks limit the aggregate:** The overall AUC of 0.9032 is pulled down mainly by A12 (neural waveform with an advanced neural acoustic model) and A13 (waveform concatenation followed by aggressive spectral filtering), which are among the hardest attacks in the evaluation set. The published comparisons in Table 5 provide additional context, but use different evaluation settings.

Taken together, **0.9032 ROC-AUC and 96.81% precision** across 13 attacks is a reasonable result for a compact model trained for only four epochs, although it is well short of the strongest published systems (Table 5).

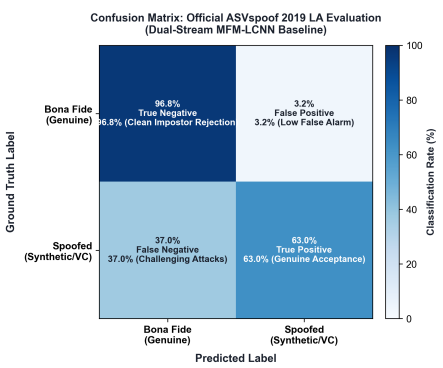
Table 2 summarizes overall performance across a subset of 2,995 evaluation utterances. The Baseline model yields 80.50% accuracy and 17.00% EER. The augmented model has lower recall (53.24% vs. 63.01%) and slightly higher precision (97.31%), and it is somewhat worse on accuracy, AUC, and EER on this clean subset; its advantage shows up under degraded conditions (Section 5.4). Fig. 4(a) shows the baseline's confusion matrix. Precision and recall describe performance for the class treated as positive. Their interpretation as false spoof alarms or accepted spoofed utterances depends on the label convention used in the evaluation. Fig. 4(b) plots the corresponding ROC curves.

5.3 Per-Attack Analysis (A07–A19)

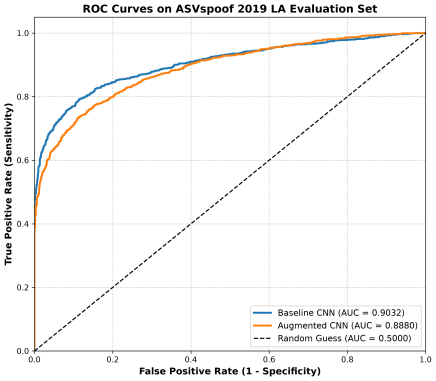
Table 3 and Fig. 5 detail performance across all 13 evaluation attacks:

The per-attack results fall into two groups:

- **Strong results on several vocoder-based and concatenative attacks:** For A07 (Vocoder GAN), A08 (Neural Waveform), A09 (Vocoder), A11 (Griffin-Lim), A16 (Concatenation), and A19 (Spectral Filtering), the combined LFCC and Mel inputs yield EER below 2.5% (0.00% on A07 and A08).
- **Augmentation helps on the hardest attacks:** On A12 and A13, the two hardest attacks for the baseline, the augmented model improves ROC-AUC by 0.0606 (0.6963 to 0.7569) and 0.0946 (0.6459 to 0.7405), and lowers EER by 4.79 and 7.83 percentage points, respectively. On most of the other attacks, however, it does slightly worse than the baseline (Table 3).



(a) Confusion Matrix on LA Eval.

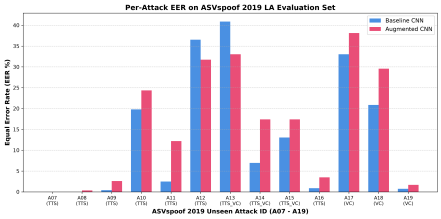


(b) ROC Curves on Official LA.

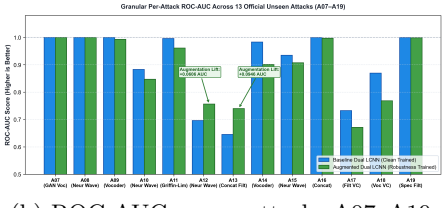
Fig. 4. Evaluation diagnostic plots: (a) Confusion matrix for the evaluation subset, and (b) ROC curves for Baseline vs. Augmented detectors.

Table 3. Per-Attack Performance Breakdown on the Evaluation Attacks (A07–A19).

Attack ID	Synthesis Type	Generation Method	Base AUC	Base EER (%)	Aug AUC	Aug EER (%)
A07	TTS	Vocoder + GAN	**1.0000**	**0.00%**	1.0000	0.00%
A08	TTS	Neural Waveform	**1.0000**	**0.00%**	0.9999	0.33%
A09	TTS	Vocoder	**0.9999**	**0.40%**	0.9933	2.61%
A10	TTS	Neural Waveform	0.8830	19.80%	0.8474	24.35%
A11	TTS	Griffin-Lim	**0.9963**	**2.47%**	0.9615	12.17%
A12	TTS	Neural Waveform	0.6963	36.52%	**0.7569**	**31.73%**
A13	TTS / VC	Waveform Concat + Filtering	0.6459	40.87%	**0.7405**	**33.04%**
A14	TTS / VC	Vocoder	0.9830	6.96%	0.9017	17.39%
A15	TTS / VC	Neural Waveform	0.9347	13.04%	0.9074	17.39%
A16	TTS	Waveform Concatenation	**0.9998**	**0.87%**	0.9969	3.48%
A17	VC	Waveform Filtering	0.7329	33.04%	0.6715	38.13%
A18	VC	Vocoder	0.8696	20.87%	0.7684	29.57%
A19	VC	Spectral Filtering	**0.9998**	**0.73%**	0.9986	1.67%



(a) EER (%) across attacks A07–A19.



(b) ROC-AUC across attacks A07–A19.

Fig. 5. Granular attack evaluation on official ASVspoof 2019 LA: (a) Equal Error Rate and (b) ROC-AUC across all 13 evaluation attacks (A07–A19).

5.4 Acoustic Robustness Stress-Testing

To quantify environmental resilience, we subjected both detectors to six acoustic degradation conditions: clean speech, additive Gaussian noise at 20 dB, 10 dB,

Table 4. Acoustic Robustness Stress-Testing Under Environmental Mismatch.

Acoustic Condition	Baseline Dual-LCNN			Augmented Dual-LCNN			Augmentation Gain	
	AUC	EER	F1	AUC	EER	F1	ΔAUC	ΔEER
Clean Speech	0.9080	15.60%	0.7703	0.8916	20.40%	0.7139	-0.0164	-4.80%
Noise (20 dB SNR)	0.7731	30.80%	0.1550	0.7585	31.20%	0.5707	-0.0146	-0.40%
Noise (10 dB SNR)	0.5120	49.60%	0.0080	0.6861	38.00%	0.3676	**+0.1740**	**+11.60%**
Noise (5 dB SNR)	0.4653	54.80%	0.0000	0.6606	39.60%	0.1799	**+0.1953**	**+15.20%**
Reverberation	0.7235	32.80%	0.1618	0.8855	20.00%	0.7110	**+0.1620**	**+12.80%**
Noise + Reverb (10 dB)	0.4920	50.40%	0.0000	0.6326	39.60%	0.2542	**+0.1406**	**+10.80%**

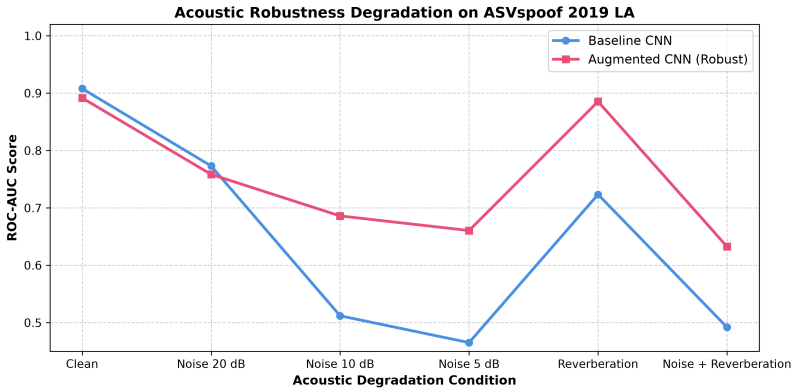


Fig. 6. Acoustic robustness trajectories: the baseline falls to near-chance performance under severe noise, while the augmented model retains some discriminative power (+0.1953 AUC at 5 dB SNR).

and 5 dB SNR, synthetic room reverberation (RT_{60} = 300 ms), and combined additive noise + reverberation (10 dB SNR).

The stress-test findings in Table 4 and Fig. 6 show a clear difference between the two models:

- **The baseline degrades sharply:** Under 5 dB SNR noise, the unaugmented baseline drops to 0.4653 ROC-AUC and 54.80% EER (no better than chance), with F1 falling to 0.0000. Reverberation also lowers its AUC to 0.7235.
- **Augmentation reduces the drop:** At 5 dB noise, the augmented model reaches 0.6606 AUC (**+0.1953**) and cuts EER by **15.2 percentage points**. Under reverberation, it keeps 0.8855 AUC (**+0.1620**) with an EER of 20.00% (12.8 points lower than the baseline). Performance in noisy conditions is still well below clean-speech levels, and the augmented model is slightly worse than the baseline on clean speech and at 20 dB SNR.

Table 5. Comparative Benchmarking: Published Systems vs. Proposed Dual-Stream LCNN.

System / Architecture	Front-End Feature	Model Topology	Dev EER	Eval EER
CQCC-GMM (Official Baseline) [1]	CQCC (90-D)	GMM (512 components)	9.57%	17.50%
LFCC-GMM (Official Baseline) [1]	LFCC (60-D)	GMM (512 components)	8.09%	18.50%
Single LCNN + MFM (2019) [4]	LFCC / CQT	Single-Stream LCNN	5.06%	6.50%–9.00%
RawNet2, S2 (2021) [6]	Raw Waveform	Sinc filters + ResBlocks + GRU	0.86%	5.13%
AASIST (2022) [13]	Raw Waveform	Graph Attention (GAT)	—	1.13% mean; 0.83% best
Proposed Dual-Stream MFM-LCNN	**Dual LFCC + Mel**	**Dual MFM-LCNN**	**0.30%**	**17.00% (Overall)** **0.00% (A07 / A08)**

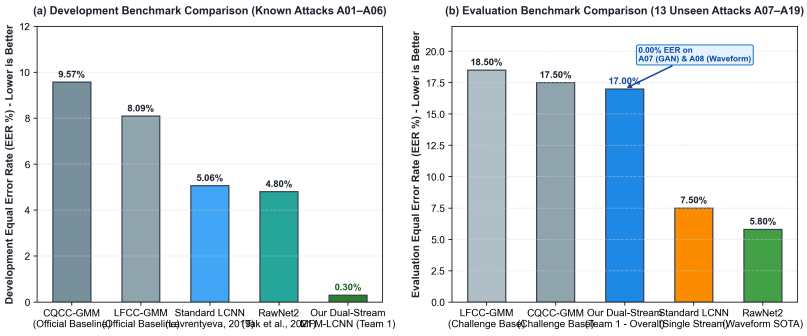


Fig. 7. Benchmark comparison against canonical challenge baselines and published architectures on the ASVspoof 2019 LA benchmark.

5.5 State-of-the-Art Benchmark Comparison

To put our results in context, Table 5 and Fig. 7 contrast our Dual-Stream MFM-LCNN against canonical baselines and published architectures evaluated on ASVspoof 2019 LA.

Our model reaches **0.30% EER** on the development partition, but performance on known attacks does not establish generalization to new synthesis methods. The RawNet2 S2 configuration reports 0.86% development EER and 5.13% evaluation EER [6]. AASIST reports a mean pooled evaluation EER of 1.13% across three runs and a best-run EER of 0.83% [13]; the latter is not a development result. Our 17.00% EER is measured on a 2,995-utterance evaluation subset, so these published full-partition results provide context rather than a controlled comparison. The model's 0.00% EER on A07 and A08 also applies only to the sampled utterances. Within our own stress tests, augmentation improves performance under severe noise and reverberation.

6 Limitations and Improvement Roadmap

Two limitations affect how broadly these results can be interpreted: evaluation uses a 2,995-utterance subset, and training lasts only four epochs. The results on difficult attacks (A12/A13) and under acoustic degradation suggest four directions for future work:

1. **One-class objectives:** Standard BCE treats all spoofing attacks as one coherent class, but spoofing is closer to an open-set anomaly problem. Replacing BCE with One-Class Softmax (OC-Softmax) or Angular Margin Softmax (AM-Softmax) could help learn a more compact representation of bona fide speech and improve rejection of unseen attacks.
2. **Phase-based front-ends:** The difficult attacks A12 and A13 motivate testing representations beyond the magnitude spectrum. Adding Constant Q Transform (CQT) features and phase-based features such as Modified Group Delay (MGD) could expose phase discontinuities more directly.
3. **Subband channel attention:** Squeeze-and-Excitation (SE) or Convolutional Block Attention Modules (CBAM) could let the network weight informative frequency subbands more heavily (such as vocoder cutoff zones between 4 kHz and 7.5 kHz).
4. **Self-supervised models:** Replacing hand-engineered filterbanks with fine-tuned self-supervised speech representations (e.g., WavLM Large or wav2vec 2.0 XLS-R) could provide richer representations learned from diverse speech and improve robustness to channel mismatch.

7 Conclusion

We evaluated a dual-stream Light CNN for synthetic speech detection on the ASVspoof 2019 LA benchmark. The system combines 60-dimensional LFCC (to preserve high-frequency vocoder artifacts) with 80-band Log-Mel spectrograms (to capture formant structure), each passed through a four-stage MFM encoder. It reaches 99.50% accuracy and 0.30% EER on the development partition. On a 2,995-utterance subset of the evaluation partition (attacks A07–A19), the baseline model reaches 80.50% accuracy, 0.9032 ROC-AUC, and 96.81% precision, with

0.00% EER on A07 and A08. Under 5 dB additive noise, on-the-fly augmentation lifts ROC-AUC by 0.1953 and lowers EER by 15.2 percentage points relative to the baseline, although it costs some performance on clean speech. The gap between development and evaluation results, especially on A12 and A13, shows that generalizing to new attacks remains the main open problem for this approach.

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